6 MAINS ELECTRICITY

The electricity that we use for heating, lighting and air conditioning in our homes is called mains electricity and is supplied to us by power stations.

In this chapter you will learn how mains electricity is brought to our homes and supplied to our appliances. You will also read about devices that protect us from electrical shocks.



▲ Figure 6.1 Most household appliances use mains electricity as their source of energy.

LEARNING OBJECTIVES

- Understand how the use of insulation, double insulation, earthing, fuses and circuit breakers protects the device or user in a range of domestic appliances
- Understand why a current in a resistor results in the electrical transfer of energy and an increase in temperature, and how this can be used in a variety of domestic contexts
- Know and use the relationship between power, current and voltage:

  power = current × voltage

  $P = I \times V$

  and apply the relationship to the selection of appropriate fuses
- Know the difference between mains electricity being alternating current (a.c.) and direct current (d.c.) being supplied by a cell or battery
- Use the relationship between energy transferred, current, voltage and time:

  energy transferred = current × voltage × time

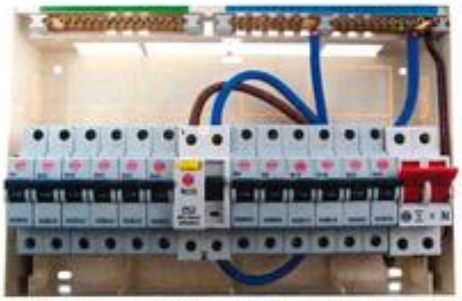
  $E = I \times V \times t$

UNITS

In this unit, you will need to use ampere (A) as the unit of current, coulomb (C) as the unit of charge, joule (J) as the unit of energy, ohm (Ω) as the unit of resistance, second (s) as the unit of time, volt (V) as the unit of voltage and watt (W) as the unit of power.

When you turn on your computer, television and most other appliances in your home the electricity you use is almost certainly going to come from the mains supply. This electrical energy usually enters our homes through an underground cable. The cable is connected to an electricity meter, which measures the amount of electrical energy used. From here, the cable is connected to a consumer unit or a fuse box, which contains fuses or circuit breakers for the various **circuits** in your home. Fuses and circuit breakers are safety devices which shut off the electricity in a circuit if the current in them becomes too large (see page 61).

Most of the wires that leave the fuse box are connected to ring main circuits that are hidden in the walls or floors around each room. Individual pieces of electrical equipment are connected to these circuits using plugs.



▲ Figure 6.2 Circuit breakers in a consumer unit

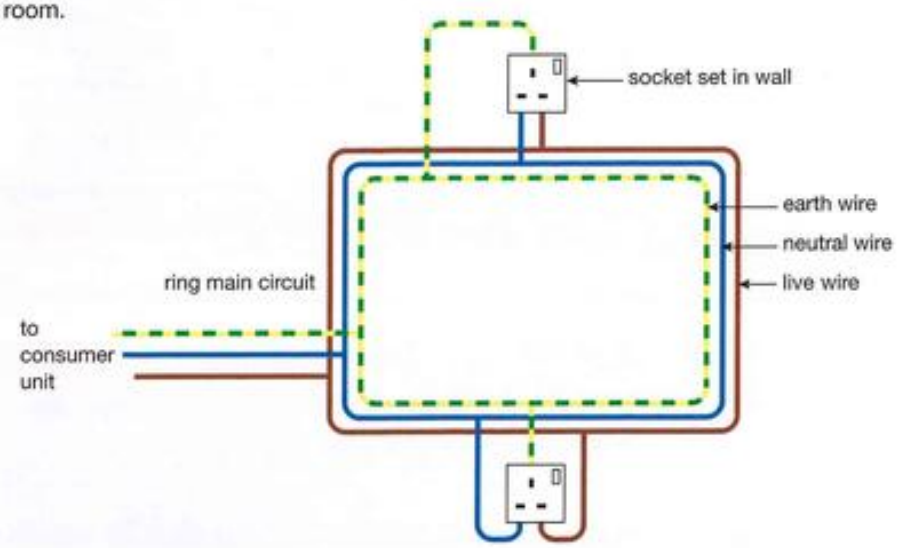
Ring main circuits usually consist of three wires – the live wire, the neutral wire and the earth wire.

The live wire provides the path along which the electrical energy from the power station travels. The neutral wire completes the circuit.

The earth wire usually has no current in it. It is there to protect you if an appliance develops a fault (see page 61). It provides a path for current to escape without passing through the user.

EXTENSION WORK

Ring main circuits provide a way of allowing several appliances in different parts of the same room to be connected to the mains using the minimum amount of wiring. Imagine how much wire would be needed if there was just one mains socket in each room.



▲ Figure 6.3 Ring main circuits help to cut down on the amount of wiring needed in a house.

Plugs and sockets in different countries look different, but the principles (rules) of electrical wiring are similar.

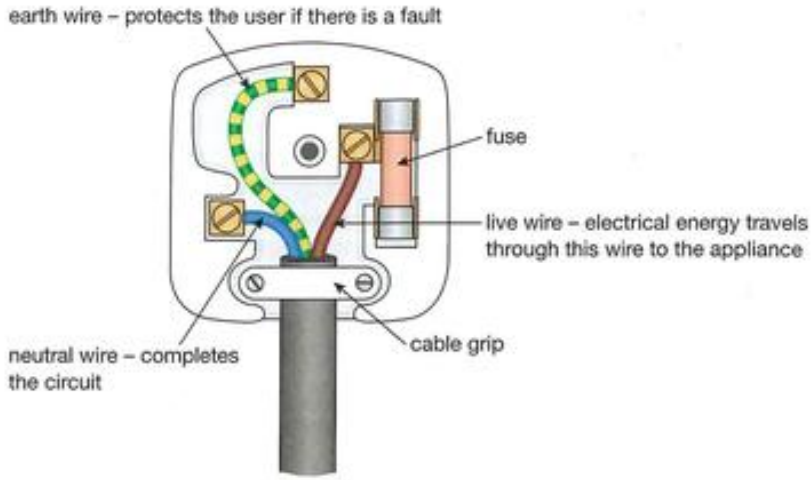




▲ Figure 6.4 Plug sockets in the UK and in Portugal

The mains electricity supplied to homes in the UK, China, India and many other countries is between 220 V and 240 V. This is a much higher **voltage** than the **cells** and batteries used in mobile electrical appliances. If you come into direct contact with mains electricity you could receive a severe electric shock, which might even be fatal. To prevent this the outer part of a plug, called the casing, is made from plastic, which is a good **insulator**.

Connections to the circuits are made via three brass pins, as the metal brass is an excellent **conductor** of electricity. Figure 6.5 shows the inside of a 3-pin plug used in the UK, but similar principles apply to all kinds of plug all over the world.



▲ Figure 6.5 A labelled plug

SAFETY DEVICES

FUSES

Many plugs contain a fuse. The fuse is usually in the form of a cylinder or cartridge, which contains a thin piece of wire made from a metal that has a low melting point. If there is too large a current in the circuit the fuse wire becomes very hot and melts. The fuse 'blows'. The circuit is now incomplete so there is no current. This prevents you getting a shock and reduces the possibility of an electrical fire. Once the fault causing the increase in current has been corrected, the blown fuse must be replaced with a new one of the same size before the appliance can be used again.



circuit symbol for a fuse

There are several sizes of fuses. The most common for domestic appliances in the UK are 3 A, 5 A and 13 A. The correct fuse for a circuit is the one that allows the correct current but blows if the current is a little larger. If the correct current in a circuit is 2 A then it should be protected with a 3 A fuse. If the correct current is 4 A then a 5 A fuse should be used. It is possible to calculate the correct size of fuse for an appliance but nowadays manufacturers provide appliances already fitted with the correct size of fuse.

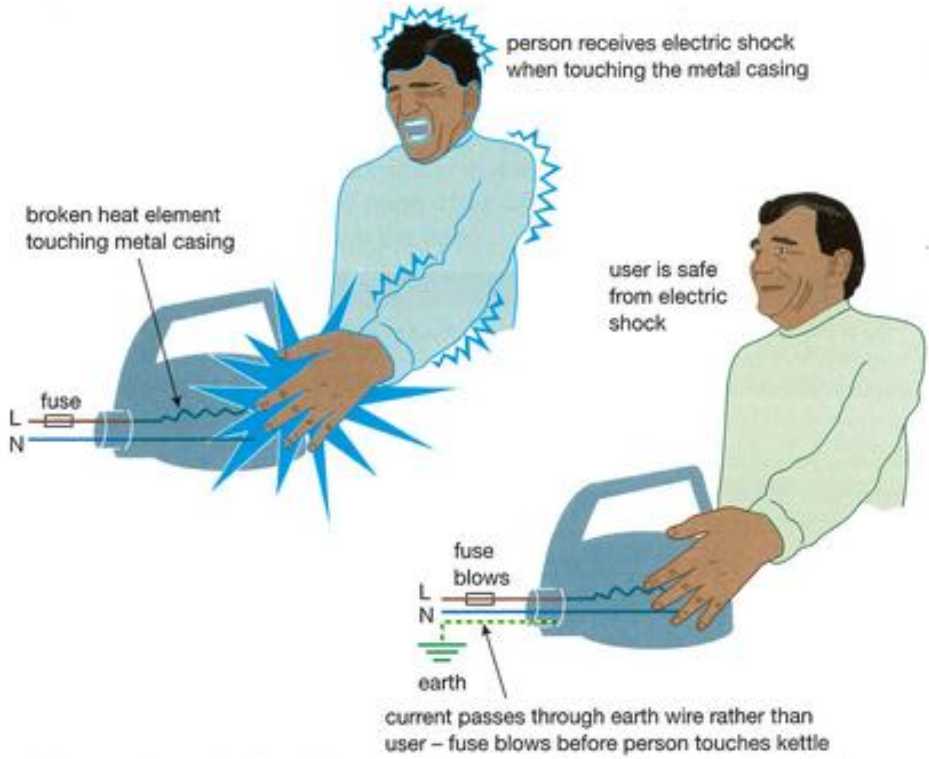


▲ Figure 6.6 Fuses are important safety devices in electrical appliances.

Modern safety devices, such as those you might find in your consumer unit, are often in the form of trip switches or circuit breakers. If too large a current flows in a circuit a switch automatically opens making the circuit incomplete. Once the fault in the circuit has been corrected, the switch is reset, usually by pressing a reset button. There is no need for the switch or circuit breaker to be replaced, as there is when fuses are used. The consumer unit shown in Figure 6.2 uses circuit breakers.

EARTH WIRES AND DOUBLE INSULATION

Many appliances have a metal casing. This should be connected to the earth wire so that if the live wire becomes damaged or breaks and comes into contact with the casing the earth wire provides a low-resistance path for the current. This current is likely to be large enough to blow the fuse and turn the circuit off. Without the earth wire anyone touching the casing of the faulty appliance would receive a severe electric shock as the current passed through them to earth (Figure 6.7).



▲ Figure 6.7 The earth wire provides protection when electrical appliances develop a fault.

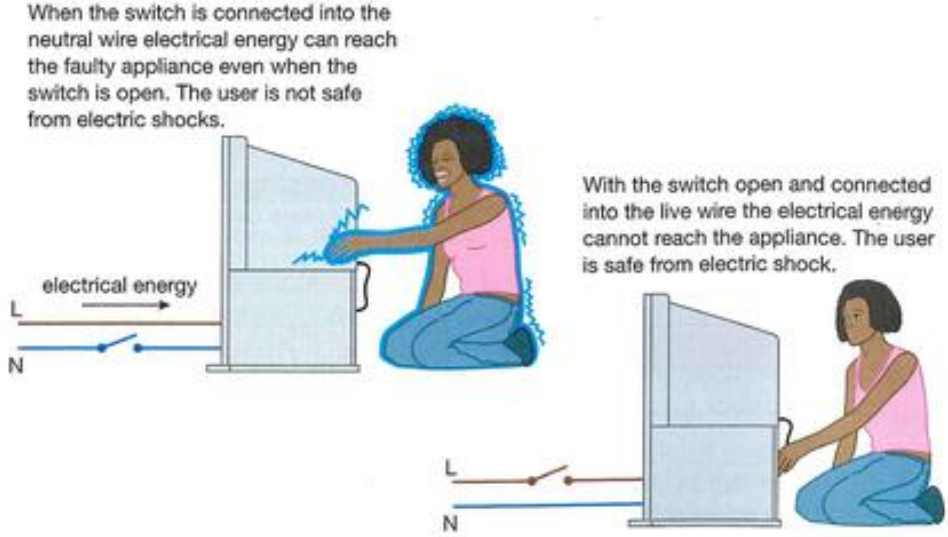


▲ Figure 6.8 This plastic kettle has double insulation which means that there is no need for an earth wire.

Some modern appliances now use casings made from an insulator such as plastic rather than from metal. If all the electrical parts of an appliance are insulated in this way, so that they cannot be touched by the user, the appliance is said to have double insulation. Appliances that have double insulation use a two-wire **flex**. There is no need for an earth wire.

SWITCHES

Switches in mains circuits should always be placed in the live wire so that when the switch is open no electrical energy can reach an appliance. If the switch is placed in the neutral wire, electrical energy can still enter a faulty appliance, and could possibly cause an electric shock (Figure 6.9).



▲ Figure 6.9 The switch in a circuit should be in the live wire.

THE HEATING EFFECT OF CURRENT



▲ Figure 6.10 The wires inside a toaster have a high resistance. They become very hot when a current passes through them.

The wiring in a house is designed to let current pass through it easily. As a result, the wires do not become warm when appliances are being used. We say that the wires have a low **resistance**. However, in some appliances, for example, kettles or toasters (Figure 6.10), we want wires (more usually called **heating elements**) to become warm. The wires of a heating element are designed to have a high resistance so that as the current passes through them energy is transferred and the element heats up. We use this heating effect of current in many different ways in our homes. You will learn more about resistance in Chapter 8.

Other common appliances that make use of the heating effect of electricity include kettles, dishwashers, electric cookers, washing machines, electric fires and hairdryers.

When current passes through the very thin wire (**filament**) of a traditional light bulb it becomes very hot and **glows** (shines) white. The bulb is transferring electrical energy to heat and light energy.

EXTENSION WORK

Incandescent bulbs like these are very inefficient. Often more than 90% of the energy transferred is lost to the surroundings as heat. As a result they are rapidly being replaced by modern halogen bulbs and LEDs, which give off much less heat and therefore waste less energy.



▲ Figure 6.11 It is the heating effect of a current that is causing this bulb to glow.

ELECTRICAL POWER

Figure 6.12 shows a 50 W **halogen light bulb**. You can also buy 70 W light bulbs. Both bulbs transfer electrical energy to heat and light. The 70 W bulb will be brighter because it transfers 70 J of electrical energy every second. The dimmer 50 W halogen bulb shown transfers only 50 J of energy every second. A 70 W bulb has a higher power rating.



▲ Figure 6.12 The dimmer 50 W bulb transfers less electrical energy to heat and light energy every second.

KEY POINT

You will learn more about voltage and current in Chapter 7.

Power is measured in joules per second or watts (W).

Devices that transfer lots of energy very quickly have their power rating expressed in kilowatts (kW).

1 kW = 1000 W

The power (P) of an appliance is related to the voltage (V) across it and the current (I) flowing through it.

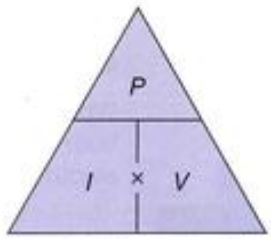
The equation is:

power, P (watts) = current, I (amps) × voltage, V (volts)

$$P = I \times V$$

HINT

If an examination question asks for the equation for calculating power, voltage or current, always give the actual equation (such as $P = I \times V$). You may not be awarded a mark if you just draw the triangle.



▲ Figure 6.13 You can use the triangle method for rearranging equations like $P = I \times V$.

EXAMPLE 1

A 230 V television takes a current of 3 A. Calculate the power of the television.

$$P = I \times V$$
$$= 3\text{ A} \times 230\text{ V}$$
$$= 690\text{ W}$$

EXAMPLE 2

Calculate the correct fuse that should be used for a 230 V, 1 kW electric hairdryer.

$$I = \frac{P}{V}$$
$$= \frac{1000\text{ W}}{230\text{ V}}$$
$$= 4.35\text{ A}$$

The correct fuse for this hairdryer is therefore a 5 A fuse.

CALCULATING THE TOTAL ENERGY TRANSFERRED BY AN APPLIANCE

The power of an appliance (P) tells you how much energy it transfers each second. This means that the total energy (E) transferred by an appliance is equal to its power multiplied by the length of time (in seconds) the appliance is being used.

energy, E (joules) = power, P (watts) × time, t (seconds)

$$E = P \times t$$

or since $P = I \times V$

$$E = I \times V \times t$$

EXAMPLE 3

Calculate the energy transferred by a 60 W bulb that is turned on for **a** 20 s, and **b** 5 min.

a $E = P \times t$

$= 60\ \text{W} \times 20\ \text{s}$

$= 1200\ \text{J}$ or $1.2\ \text{kJ}$

b $E = P \times t$

$= 60\ \text{W} \times 5 \times 60\ \text{s}$

$= 18\,000\ \text{J}$ or $18\ \text{kJ}$

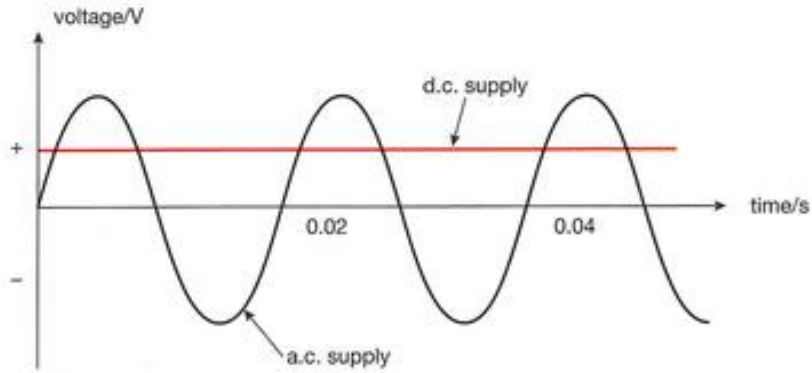
ALTERNATING CURRENT AND DIRECT CURRENT

If we could see the current or voltage from the mains it would appear to be very strange. Its value increases and then decreases and then does the same again but in the opposite direction. If we could draw these changes as a graph they would look like a wave.

This happens because of the way in which the electricity is generated at the power station. (see Chapter 17). A current or voltage that behaves like this is called an alternating current (a.c.) or **alternating voltage**. This is very different to the currents and voltages we get from batteries and cells.

Cells and batteries provide currents and voltages that are always in the same direction and have the same value. This is called direct current (d.c.) or direct voltage. If we drew this as a graph it would be a straight horizontal line.

Figure 6.14 shows how the voltage of an a.c. supply compares with the voltage of a d.c. supply.



▲ Figure 6.14 How the voltage of an a.c. supply compares with that of a d.c. supply

CHAPTER QUESTIONS

More questions on domestic electricity can be found at the end of Unit 2 on page 93.

SKILLS PROBLEM SOLVING (7)

1 a There is a current of 0.25 A in a bulb when a voltage of 12 V is applied across it. Calculate the power of the bulb.

(8) b Calculate the voltage that is being applied across a 10 W bulb with a current of 0.2 A.

c Calculate the current in a 60 W bulb if the voltage across it is 230 V.

(7) d How much energy is transferred if a 100 W bulb is left on for 5 hours?

SKILLS CRITICAL THINKING

2 An electric kettle is marked '230 V, 1.5 kW'.

(7) a Explain what these numbers mean.

SKILLS PROBLEM SOLVING (6) b Calculate the correct fuse that should be used.

SKILLS REASONING c Explain why a 230 V, 100 W bulb glows more brightly than a 230 V, 60 W bulb when both are connected to the mains supply.

SKILLS CRITICAL THINKING (5)

3 a Give one advantage of using a circuit breaker rather than a wire or cartridge fuse.

b Why is the switch for an appliance always placed in the live wire?

c What is meant by the sentence 'The hairdryer has double insulation'?

SKILLS DECISION MAKING, CREATIVITY (6)

4 Think of a room in your house where there are lots of electrical appliances. Make a list of them. Now organise your list so that the appliances that you think have the highest power rating are at the top of your list and those with the lowest are at the bottom. How could you discover if your guesses are correct?